EPSR-test-ex4 Feedback factor (feedback 0.8)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the `feedback` factor of the `epsr` engine [1,2]: the confidence placed in the data when the empirical potential is derived, entering the inversion of the neutron weights matrix. Changing it from the value stored with the run forces the inverse weights to be recalculated, a code path not taken by the base case.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `get_wts` (detects `abs(feedback - feedbackold)` > 0.001), `invert_cwt` (inverse weights with feedback)
- `epsr_standalone_rev3.f`: `ptgnsq` (EP refinement with the new inverse weights), `runepsr17`
- `read_and_save_files.f`: `read_input_file`, `save_files_others`

Rust (port, rust/epsr-core/src/):

- `weights.rs`: `State::get_wts`, `invert_cwt`
- `potential.rs`: `State::ptgnsq`; `epsr.rs`: `Epsr::setup`, `Epsr::iterate`
- `symbols.rs` / `epsr.rs`: `.inp` reading and saving

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [3] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [4]. With `feedback` f the structure factors used to update the EP are formed from a fraction f of the data and 1 - f of the simulation; f is lowered from 0.9 to 0.8.

Inputs

Copy of EPSR-test with only `feedback 0.8` (instead of 0.9) in `200DMSO400Water.EPSR.inp`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Two chained passes: in pass 1 `get_wts` compares the new factor with the value stored in the `.inpa` accumulator file (`feedbackold` 0.9) and rebuilds the inverse weights matrix; the `.inpa` written then holds 0.8, so pass 2 reuses the stored inverse. The written `.inp` keeps `feedback 0.8`. All 27 output files, including `.pcof` with the refined EP coefficients and `delfit.dat`, are identical after both passes, which requires the same double-precision matrix inversion and the same order of `ran1` draws.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.